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COMMISSION DELEGATED REGULATION (EU) 2017/79

of 12 September 2016

establishing detailed technical requirements and test procedures for the EC type-approval of motor vehicles with respect to their 112-based eCall in-vehicles systems, of 112-based eCall in-vehicle separate technical units and components and supplementing and amending Regulation (EU) 2015/758 of the European Parliament and of the Council with regard to the exemptions and applicable standards

(Text with EEA relevance)

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(Text with EEA relevance)

Article 1

Subject matter

This Regulation establishes detailed technical requirements and test procedures for the EC type-approval of the vehicles referred to in Article 2 of Regulation (EU) 2015/758 in respect of their 112-based eCall in-vehicle systems and of 112-based eCall in-vehicle separate technical units ('STUs') and components.

Article 2

Classes of vehicles exempted from the requirement to be equipped with a 112-based eCall in-vehicle system

The classes of vehicles which for technical reasons cannot be fitted with an appropriate eCall triggering mechanism and for that reason are exempted from the requirement to be equipped with a 112-based eCall in-vehicle system are listed in Annex IX.

Article 3

Multi-stage approval of special purpose vehicles

In case of multi-stage type-approval of the special purpose vehicles defined in points 5.1 and 5.5 of part A of Annex II to Directive 2007/46/EC, the type-approval granted at a previous stage in respect of the installation of a 112-based eCall in-vehicle system in the (base) vehicle shall remain valid, provided that the 112-based eCall in-vehicle system and the relevant sensors are not modified.

Article 4

Definitions

For the purposes of this Regulation the following definitions shall apply:

(1) 'vehicle type with regard to the installation of a 112-based eCall in-vehicle system' means motor vehicles that do not differ in such essential respects as the characteristics of the integration within the vehicle as well as the functionality and capability of essential hardware deploying an in-vehicle emergency call.

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(2) 'type of 112-based eCall in-vehicle STU' means a combination of specific hardware which does not differ in such essential respects as the characteristics, functionality and capability of deploying an in-vehicle emergency call when installed in a motor vehicle.

(3) 'type of 112-based eCall in-vehicle system component' means specific hardware which does not differ in such essential respects as the characteristics, functionality and capability of facilitating the deployment of an in-vehicle emergency call when integrated in a 112-based eCall in-vehicle STU or 112-based eCall in-vehicle system.

(4) 'representative arrangement of parts' means all parts required by the 112-based eCall in-vehicle system to successfully populate and transmit in an in-vehicle emergency call the minimum set of data referred to in the standard EN 15722:2015 'Intelligent transport systems — eSafety — eCall minimum set of data' ('MSD') including the control module, the power source, the mobile network communication module, the Global Navigation Satellite System receiver and the external Global Navigation Satellite System antenna and their connectors and wiring;

(5) 'control module' means a component of the e-Call in-vehicle system designed to ensure the combined functioning of all modules, components and features of the system;

(6) 'power source' means the component that supplies power to the 112-based e-Call in-vehicle system, including a back-up source if fitted, which feeds the system after the test referred to in point 2.3 of Annex I;

(7) 'eCall log file' means any record generated at the moment of an automatic or manual eCall activation which is stored within the internal memory of the 112-based eCall in-vehicle system and consists only of the MSD;

(8) 'Global Navigation Satellite System' ('GNSS') means an infrastructure composed of a constellation of satellites and a network of ground stations, which provides accurate timing and geolocation information to users having an appropriate receiver;

(9) 'Satellite-Based Augmentation System' ('SBAS') means a regional navigation satellite system for monitoring and correcting signals emitted by existing global satellite navigation systems, giving the users better performance in terms of accuracy and integrity;

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(10) 'cold start mode' means the condition of a GNSS receiver when position, velocity, time, almanac and ephemeris data are not stored in the receiver and therefore the navigation solution is to be calculated by means of a full sky search;

(11) 'up-to-date location' means the last known vehicle position determined at the latest moment possible before generation of the MSD;

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(12) 'test eCall' means an eCall for testing purposes, which can be clearly distinguished from a real eCall or which does not reach the Public Safety answering Point ('PSAP').

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Article 5

Requirements and test procedures for EC type-approval of motor vehicles with regard to the installation of 112-based eCall in-vehicle systems

1. EC type-approval of a vehicle with regard to the installation of a 112-based eCall in-vehicle system shall be subject to the vehicle and its system passing the tests laid down in Annexes I to VIII and complying with the relevant requirements laid down in those Annexes.

2. Where the motor vehicle is fitted with a type of 112-based eCall in-vehicle STU that has been type-approved in accordance with Article 7, the vehicle and its system shall have to pass the tests laid down in Annexes II, III and V and to comply with all relevant requirements laid down in those Annexes.

3. Where the 112-based eCall in-vehicle system of the motor vehicle comprises one or more components that have been type-approved in accordance with Article 6, the motor vehicle and its system shall have to pass the tests laid down in Annexes I to VIII and to comply with all relevant requirements laid down in those Annexes. The assessment of whether the system complies with those requirements may however partly be based on the results of the tests referred to in Article 6(3).

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4. For the purpose of extending the EC type-approval granted in accordance with paragraph (1) before 1 January 2027, the technical service may exempt 112-based eCall in-vehicle system from full-scale impact test as specified in Annex II and from the subsequent audio equipment test as specified in Annex III. The changes of the modified 112-based eCall in-vehicle system as compared to the originally approved system shall be documented and explained by the manufacturer to the technical service and the type-approval authority:

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(a) Where the communication part is modified without impacting other components of the 112-based eCall in-vehicle system and a vehicle crash-test is conducted for other purposes, the 112-based eCall in-vehicle system shall be included and the full-scale impact test as specified in Annex II and the subsequent audio equipment test as specified in Annex III shall be conducted.

(b) Where the modification of the communication part of a 112-based eCall in-vehicle system has an impact on its other parts, the full-scale impact test as specified in Annex II and the subsequent audio equipment test as specified in Annex III shall be conducted.

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Article 6

Requirements and test procedures for EC type-approval of 112-based eCall in-vehicle system components

1. EC type-approval of a 112-based eCall in-vehicle system component shall be subject to the component passing the tests laid down in Annex I and complying with the relevant requirements in that Annex.

2. For the purposes of paragraph 1, only the verification procedure for components laid down in point 2.8 of Annex I shall apply after the individual parts are subjected to the test referred to in point 2.3 of this Annex.

3. Upon request of the manufacturer, a component may additionally be tested by the technical service for compliance with the requirements set out in Annexes IV, VI and VII that are relevant to the functionalities of the component. Compliance with those requirements shall be indicated on the type-approval certificate issued in accordance with Article 3(3) of Commission Implementing Regulation (EU) 2017/78 ([1]).

Article 7

Requirements and test procedures for EC type-approval of 112-based eCall in-vehicle STUs

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1. EC type-approval of a 112-based eCall in-vehicle STU shall be subject to the STU passing the tests laid down in Annexes I, IV, VI, VII and VIII and complying with the relevant requirements laid down in those Annexes. In case the STU is fitted with a back-up power source, it shall comply with the requirements and be subject to the test procedure laid down in Annex X.

([1]) Commission Implementing Regulation (EU) 2017/78 of 15 July 2016 establishing administrative provisions for the EC type-approval of motor vehicles with respect to their 112-based eCall in-vehicle systems and uniform conditions for the implementation of Regulation (EU) 2015/758 of the European Parliament and of the Council with regard to the privacy and data protection of users of such systems (see page 26 of this Official Journal).

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2. Where the 112-based eCall in-vehicle STU comprises one or more components that have been type-approved in accordance with Article 6, the STU shall have to pass the tests laid down in Annexes I, IV, VI, VII and VIII and to comply with all relevant requirements laid down in those Annexes. The assessment of whether the STU complies with those requirements may however partly be based on the results of the test referred to in Article 6(3).

Article 8

Obligations of the Member States

Member States shall refuse to grant EC type-approval for new types of motor vehicles that do not comply with the requirements set out in this Regulation.

Article 9

Amendments to Regulation (EU) 2015/758

The second subparagraph of Article 5(8) of Regulation (EU) 2015/758 is replaced by the following:

‘The technical requirements and tests referred to in the first subparagraph shall be based on the requirements set out in paragraphs 2 to 7 and on the available standards relating to eCall, where applicable, including:

(a) EN 16072:2015 “Intelligent transport systems — eSafety — Pan-European eCall operating requirements”;

(b) EN 16062:2015 “Intelligent transport systems — eSafety — eCall high level application requirements (HLAR)”;

(c) EN 16454:2015 “Intelligent transport systems — ESafety — Ecall end to end conformance testing”;

(d) EN 15722:2015 “Intelligent transport systems — eSafety — eCall minimum set of data (MSD)”;

(e) EN 16102:2011 “Intelligent transport systems — eCall — Operating requirements for third party support”;

(f) any additional European standards relating to the eCall system adopted in conformity with the procedures laid down in Regulation (EU) No 1025/2012 of the European Parliament and of the Council (*), or Regulations of the United Nations Economic Commission for Europe (UNECE Regulations) relating to eCall systems to which the Union has acceded.

(*) Regulation (EU) No 1025/2012 of the European Parliament and of the Council of 25 October 2012 on European standardisation, amending Council Directives 89/686/EEC and 93/15/EEC and Directives 94/9/EC, 94/25/EC, 95/16/EC, 97/23/EC, 98/34/EC, 2004/22/EC, 2007/23/EC, 2009/23/EC and 2009/105/EC of the European Parliament and of the Council and repealing Council Decision 87/95/EEC and Decision No 1673/2006/EC of the European Parliament and of the Council (OJ L 316, 14.11.2012, p. 12).’

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Article 10

Entry into force and application

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

It shall apply from 31 March 2018.

This Regulation shall be binding in its entirety and directly applicable in all Member States.

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ANNEX I

Technical requirements and procedures for testing the resistance of eCall in-vehicle systems to severe crashes (high-severity deceleration test)

1. Requirements

1.1. Performance requirements

1.1.1. The high-severity deceleration test of eCall in-vehicle systems, STUs and components, carried out in accordance with point 2, shall be considered satisfactory if the following requirements are demonstrated post-deceleration/acceleration event.

1.1.2. MSD emission and encoding: The eCall system or representative arrangement shall be able to successfully transmit an MSD to a PSAP test point.

1.1.3. Incident time determination: The eCall system or representative arrangement shall be able to determine an up-to-date timestamp for an eCall incident.

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1.1.4. Position determination: The eCall system or representative arrangement shall be able to determine accurately the up-to-date vehicle location, including two recent vehicle locations before the generation of the data for the MSD.

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1.1.5. Mobile network connectivity: The eCall system or representative arrangement shall be able to connect to and transmit data via the mobile network.

2. Test procedure

2.1. Purpose of the high-severity deceleration test procedure

The purpose of this test is to verify the sustained functionality of the 112-based eCall system after being subjected to inertial loads which may occur during a severe vehicle crash.

2.2. The following tests shall be performed on a representative arrangement of parts (without a vehicle body).

2.2.1. A representative arrangement shall include all parts required by the eCall system to successfully populate and transmit the MSD in an eCall.

2.2.2. This shall include the control module and the power source and any other parts required to perform the test eCall.

2.2.3. This shall include the external antenna for mobile communication.

2.2.4. The wiring harness may be represented only by the relevant connectors (connected to the tested components) and a length of wire. The length of the wiring harness and its eventual fixation can be decided by the manufacturer in agreement with the technical service referred to in Article 3(31) of Directive 2007/46/EC so that it is representative for the different installation configurations of the eCall system. ►M1 Any such agreement shall be documented in the test report. ◄

2.3. Deceleration/acceleration procedure

2.3.1. The following conditions shall apply:

(a) The test shall be conducted at an ambient temperature of 20 ± 10 °C.

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(b) At the beginning of the test, the power supply shall be charged sufficiently to allow performing the subsequent verification tests.

2.3.2. The tested parts shall be connected to the test fixture by the intended mountings provided for the purpose of attaching them to a vehicle. If the intended mountings of the power source are specifically designed to break in order to release the power source in an impact event, they shall not be included in the test. The technical service shall verify that such release in a real-life high-severity crash event shall not impair the functionality of the system (e.g. no disconnection from the power source).

2.3.3. If additional brackets or fixtures are used as part of the deceleration/acceleration facility, these shall provide a sufficiently rigid connection to the deceleration/acceleration facility to not affect the outcome of the test.

2.3.4. The eCall system shall be decelerated or accelerated in compliance with the pulse corridor that is specified in the Table and Figure. The acceleration/deceleration shall be measured at a rigid part of the deceleration/acceleration facility and filtered at CFC-60.

2.3.5. The test pulse shall be within the minimum and maximum values as specified in the Table. The maximum velocity change ΔV shall be 70 km/h [+ 0/– 2 km/h]. However, if with the agreement of the manufacturer, the test was performed at a higher acceleration or deceleration level, a higher ΔV and/or longer duration the test shall be considered satisfactory.

2.3.6. The parts referred to in point 2.2 shall be tested in a worst case configuration. Their position and orientation on the sled shall correspond to the installation recommendations of the manufacturer and shall be indicated in the type-approval certificate issued under Implementing Regulation (EU) 2017/78.

2.3.7. Description of the test pulse

Figure

Minimum and maximum curve of the test pulse (pulse corridor)

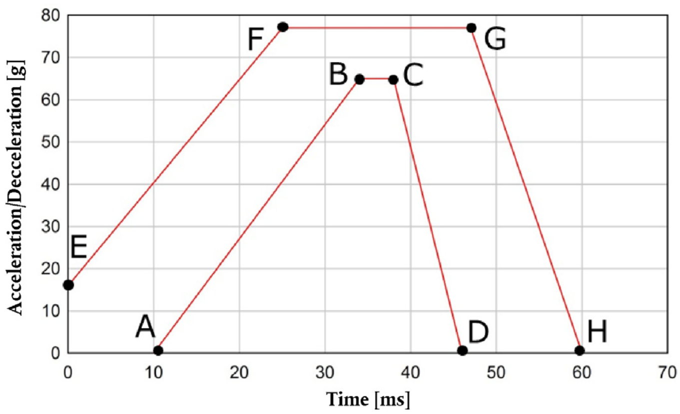

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Table

Acceleration/deceleration values of the minimum and maximum curve of the test pulse

Point	Time (ms)	Acceleration/Deceleration (g)
A	10	0
B	34	65
C	38	65
D	46	0
E	0	16
F	25	77
G	47	77
H	60	0

2.4. Verification procedure

2.4.1. Verify that no cable connectors were unplugged during the event.

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2.4.2. The performance requirements shall be verified by performing a test eCall using the power source subjected to the high-severity deceleration.

2.4.3. Before performing the test eCall, ensure that:

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(a) the eCall system receives (real or simulated) GNSS signals to an extent representative of open sky conditions;

(b) the eCall system has had sufficient time in a powered state to achieve a GNSS position fix;

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(c) one of the connection procedures defined in point 2.7, as agreed between the technical service and the manufacturer, shall be applied for any test eCall;

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(d) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(e) a false eCall to a genuine PSAP cannot be made over the live network; and

(f) if applicable, the TPS system is deactivated or will automatically switch to the 112-based system.

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2.4.4. Perform a test eCall by applying a trigger according to the instructions of the manufacturer.

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2.4.5. Verify each of the following items:

(a) Verify that an MSD was received by the PSAP test point. This shall be verified by a record of the PSAP test point showing that an MSD emitted from the eCall system following the trigger was received and successfully decoded. If the MSD decoding failed at redundancy version MSD rv0 but was successful at a higher redundancy version or in robust modulator mode, as defined in ETSI/TS 126 267, this is acceptable.

(b) Verify that the MSD contained an up-to-date timestamp. This shall be verified by a test record showing that the timestamp contained in the MSD received by the PSAP test point does not deviate from the exact recorded time of the trigger activation by more than 60 seconds. The transmission may be repeated if the eCall system failed to achieve a GNSS position fix before the test.

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(c) Verify that the MSD contained an accurate, up-to-date location. This shall be verified in accordance with the Positioning Test Procedure as specified in point 2.5 by a test record showing that the deviation between IVS location and true location, d_IVS, is less than 150 metres and the confidence bit transmitted to the PSAP test point indicates 'position can be trusted'.

(d) Verify that the MSD contained the two recent locations before the generation of the data for the MSD. This shall be verified by a record of the PSAP test point showing that it received the 'recent-VehicleLocationN1' and 'recentVehicleLocationN2'.

2.4.6. Clear down the test eCall using the appropriate PSAP test point command (e.g. hang up).

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2.5. Positioning test procedure

2.5.1. The sustained functionality of the GNSS components shall be verified by comparing the location input and the location output of the system.

2.5.2. The 'IVS location' (φ_{IVS}, λ_{IVS}) shall be: The location contained in an MSD transmitted to a PSAP test point while the GNSS antenna is in open sky conditions (real or simulated).

2.5.3. The 'true location' (φ_{true}, λ_{true}) shall be:

(a) the actual location of the GNSS antenna (known location or determined with another means than the eCall system), when using real GNSS signals; or

(b) the simulated location, when using simulated GNSS signals.

2.5.4. The deviation between IVS location and true location, d_{IVS} shall be calculated using the following equations:

$$\Delta\varphi = \varphi_{IVS} - \varphi_{true}$$

$$\Delta\lambda = \lambda_{IVS} - \lambda_{true}$$

$$\varphi_m = \frac{\varphi_{IVS} + \varphi_{true}}{2}$$

$$d_{IVS} = R\sqrt{(\Delta\varphi)^2 + (\cos(\varphi_m)\Delta\lambda)^2}$$

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where:

$\Delta\varphi$: Difference in latitude (in radian)

$\Delta\lambda$: Difference in longitude (in radian)

Note: $1° = \frac{\pi}{180}$ rad; 1 mas = $4{,}8481368 \cdot 10^{-9}$ rad

φ_m: Mean latitude (in unit suitable for the cosine calculation)

R: Radius of the earth (mean) = 6 371 009 metres

2.5.5. The positioning test procedure may be repeated if the eCall system failed to achieve a GNSS position fix before the test.

2.6. Antenna test procedure

2.6.1. If the connection procedure applied for the test call did not make use of over-the-air data transmission, the sustained functionality of the mobile network antenna shall be verified by checking the antenna tuning status after the deceleration event according to the following procedure.

2.6.2. Measure the voltage standing wave ratio,, of the external mobile network antenna after the deceleration event at a frequency within the antenna's specified frequency band.

2.6.2.1. The measurement shall be performed with a power meter, antenna analyser or SWR meter as close as possible to the antenna feed point.

2.6.2.2. If a power meter is used, shall be calculated using the following equation:

$$VSWR = \frac{\sqrt{P_f} + \sqrt{P_r}}{\sqrt{P_f} - \sqrt{P_r}}$$

where:

P_f: Forward measured power

P_r: Reverse/reflected measured power

2.6.3. Verify that satisfies the specifications prescribed by the manufacturer for new antennas.

2.7. Connection procedures

2.7.1. Simulated Mobile Network Procedure

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2.7.1.1. It shall be ensured that an emergency call emitted by the 112-based system will be performed over-the-air via a non-public (i.e. simulated) mobile network and routed to the dedicated PSAP test point.

2.7.1.2. The dedicated PSAP test point during the test procedures shall be a PSAP simulator under the control of the technical service, compliant with the applicable EN standards and certified in accordance with EN 16454 and EN 17240. It shall be equipped with an audio interface to allow voice communication tests.

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2.7.1.3. If applicable, it shall be ensured that a TS11 call emitted by the TPS system will be performed over-the-air via a non-public (i.e. simulated) mobile network and routed to the TPSP test point.

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2.7.1.4. The TPSP test point shall be a dedicated TPSP answering point simulator under the control of the technical service or a genuine TPSP answering point (permission by TPSP required).

2.7.1.5. Mobile network coverage of at least – 99 dBm or equivalent is recommended for this procedure.

2.7.2. Public Mobile Network Procedure

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2.7.2.1. It shall be ensured that a regular call to a long number is emitted by the 112-based eCall in-vehicle system (instead of an emergency call) and is performed over-the-air via a public mobile network and routed to the dedicated PSAP test point.

In case where this procedure is technically not possible, an emergency call emitted by the 112-based eCall in-vehicle system via a public mobile network in packet-switched domain can be used instead, if agreed by the manufacturer or technical service with the genuine PSAP or if a technical solution has been set up to route the eCall to a dedicated PSAP test point.

2.7.2.2. The dedicated PSAP test point during the test procedures shall be a PSAP simulator under the control of the technical service, compliant with the applicable EN standards and certified in accordance with EN 16454 and EN 17240. It shall be equipped with an audio interface to allow voice communication tests.

2.7.2.3. If applicable, it shall be ensured that a regular call emitted by the TPS system will be performed over-the-air via a public mobile network and routed to the TPSP test point.

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2.7.2.4. The TPSP test point shall be a dedicated TPSP answering point simulator under the control of the technical service or a genuine TPSP answering point (permission by TPSP required).

2.7.2.5. Mobile network coverage of at least – 99 dBm or equivalent is recommended for this procedure.

2.7.3. Wired Transmission Procedure

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2.7.3.1. It shall be ensured that an emergency call emitted by the 112-based system will only be performed via a wired connection with a dedicated network simulator (bypassing any mobile network antenna) and routed to the dedicated PSAP test point.

2.7.3.2. The dedicated PSAP test point during the test procedures shall be a PSAP simulator under the control of the technical service, compliant with the applicable EN standards and certified in accordance with EN 16454 and EN 17240. It shall be equipped with an audio interface to allow voice communication tests.

2.7.3.3. If applicable, it shall be ensured that a regular call emitted by the TPS system will be performed via a wired connection with a dedicated network simulator (bypassing any mobile network antenna) and routed to the dedicated TPSP test point.

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2.7.3.4. The TPSP test point shall be a dedicated TPSP answering point simulator under the control of the technical service or a genuine TPSP answering point (permission by TPSP required).

2.8. Verification procedures for components

2.8.1. ►M1 These procedures shall apply for the purposes of type-approval of a 112-based eCall in-vehicle system component in accordance with Article 6 of this Regulation. ◄

2.8.1.1. These procedures shall apply after the individual parts are subjected to the deceleration test under point 2.3 of this Annex.

2.8.2. Control module including its connectors and wire harness as described in point 2.2.4 of this Annex.

2.8.2.1. Verify that no cable connectors are unplugged during the event.

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2.8.2.2. The performance requirements shall be verified by performing a test eCall according to paragraphs 2.4.3 to 2.4.6.

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2.8.3. Mobile network antenna including its connectors and wire harness as described in point 2.2.4 of this Annex

2.8.3.1. Verify that no cable connectors were unplugged during the event.

2.8.3.2. Measure the voltage standing wave ratio, VSWR, of the external mobile network antenna after the deceleration event at a frequency within the antenna's specified frequency band.

2.8.3.3. The measurement shall be performed with a power meter, antenna analyser or SWR meter as close as possible to the antenna feed point.

2.8.3.4. If a power meter is used, VSWR shall be calculated using the following equation:

$$VSWR = \frac{\sqrt{P_f} + \sqrt{P_r}}{\sqrt{P_f} - \sqrt{P_r}}$$

where:

P_f: Forward measured power

P_r: Reverse/reflected measured power

2.8.3.5. Verify that VSWR satisfies the specifications prescribed by the manufacturer for new antennas.

2.8.4. Power supply (if not part of the control module) including its connectors and wire harness as described in point 2.2.4 of this Annex

2.8.4.1. Verify that no cable connectors are unplugged during the event.

2.8.4.2. Measure if the voltage corresponds to the manufacturer's specification.

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ANNEX II

Full-scale impact test assessment

1. Requirements

1.1. Performance requirements

1.1.1. The full-scale impact assessment of vehicles with eCall in-vehicle systems installed, carried out in accordance with point 2, shall be considered satisfactory if the following requirements are demonstrated post-impact.

1.1.2. Automatic triggering: The eCall system shall automatically initiate an eCall after an impact in accordance with UN Regulation No 94 (Annex 3) as well as UN Regulation No 95 (Annex 4), as applicable.

1.1.3. Call status indication: The eCall system shall inform the occupants about the current status of the eCall (status indicator) using a visual and/or audible signal.

1.1.4. MSD emission and encoding: The eCall system shall be able to successfully transmit an MSD to a PSAP test point via the mobile network.

1.1.5. Vehicle-specific data determination: The eCall system shall be able to populate accurately the mandatory vehicle-specific data fields of the MSD.

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1.1.6. Position determination: The eCall system shall be able to determine accurately the up-to-date vehicle location, including two recent vehicle locations before the generation of the data for the MSD.

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2. Test procedure

2.1. Purpose of the full-scale impact test procedure

The purpose of this test is to verify the automatic triggering function and the sustained functionality of the 112-based eCall in-vehicle system in vehicles that are subjected to a frontal impact or a side impact.

2.2. The following tests shall be performed on a vehicle with an eCall in-vehicle system installed.

2.3. Impact test procedure

2.3.1. Impact tests shall be carried out in accordance with the tests defined in UN Regulation No 94, Annex 3 for frontal impact as well as UN Regulation No 95, Annex 4 for side impact, as applicable.

2.3.2. The test conditions defined in UN Regulation No 94 or UN Regulation No 95 shall apply.

2.3.3. Before performing the impact tests, ensure that:

(a) the in-vehicle power source, if installed for the test, is charged according to the specifications of the manufacturer at the beginning of the test to allow performing the subsequent verification tests;

(b) the automatic eCall is enabled and armed and that the vehicle ignition or master control switch is activated;

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(c) one of the connection procedures defined in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

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(d) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(e) a false eCall to a genuine PSAP cannot be made over the live network; and

(f) if applicable, the TPS system is deactivated or will automatically switch to the 112-based system.

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2.4. Power-Supply Test procedure

The test procedure described in points 2.4.1 to 2.4.6. shall apply where Annex X is applicable and any of the following conditions is met:

(a) the eCall STU was not subject of the test laid down in Annex X,

(b) the eCall STU was subject of the test laid down in Annex X but the back-up power supply in the vehicle is shared with other devices.

2.4.1. If the automatic eCall was terminated, trigger a manual test eCall.

2.4.2. Read out any text for at least 5 minutes at the PSAP test point. Alternatively, the test method referred to in point 2.6.2. of Annex III can be performed at this step if the duration of 5 minutes is not exceeded.

2.4.3. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

2.4.4. Wait for 56 minutes after the call was ended.

2.4.5. Initiate a call from the PSAP test point to the eCall in-vehicle system.

2.4.6. If the call is automatically accepted, read out any text for at least 5 minutes at the PSAP test point, otherwise the test is finished.

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2.5. Positioning test procedure

The positioning test procedure defined in point 2.5 of Annex I to this Regulation shall apply.

2.6. Antenna test procedure

2.6.1. If the connection procedure applied for the test call did not make use of over-the-air data transmission (point 2.7.3 of Annex I to this Regulation), the sustained functionality of the mobile network antenna shall be verified by checking the antenna tuning status after the full-scale impact test according to the procedure defined in point 2.6 of Annex I to this Regulation. In addition, it shall be verified that no wire breakage or short-circuit of the antenna feed line occurred by checking the electrical resistance between the end points of the wire and between the wire and vehicle ground.

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3. Verification procedure

3.1. Verification of the Minimum Set of Data (MSD)

3.1.1. Verify each of the following items in at least one of the test eCalls:

(a) Verify that an eCall was triggered automatically by the full-scale impact event. This shall be verified by a record of the PSAP test point showing that it received an eCall following the impact event and that the MSD control indicator was set to 'automatically initiated eCall'.

(b) Verify that the eCall status indicator indicated an eCall sequence following the automatic or manual trigger. This shall be verified by a record showing that an indication sequence was performed on all sensory channels specified in the manufacturer's documentation (visual and/or audible).

(c) Verify that an MSD was received by the PSAP test point. This shall be verified by a record of the PSAP test point showing that an MSD emitted from the vehicle following the automatic trigger was received and successfully decoded.

(d) Verify that the MSD contained accurate vehicle-specific data. This shall be verified by a record of the dedicated PSAP test point showing that the information transmitted in the fields regarding vehicle type, vehicle identification number (VIN) and vehicle propulsion storage type does not deviate from the information specified in the type-approval application.

(e) Verify that the MSD contained an accurate, up-to-date location. This shall be verified in accordance with the Positioning Test Procedure as defined in point 2.5 of Annex I by a test record showing that the deviation between IVS location and true location, d_IVS, is less than 150 metres and the confidence bit transmitted to the PSAP test point indicates 'position can be trusted'. If no GNSS signals are available at the impact test location, the vehicle can be moved to an appropriate location before performing the test eCall.

(f) Verify that the MSD contained the two recent locations before the generation of the data for the MSD. This shall be verified by a record of the PSAP test point showing that it received the 'recentVehicle-LocationN1' and 'recentVehicleLocationN2'.

(g) Verify that the MSD contained an up-to-date timestamp. This shall be verified by a test record showing that the timestamp contained in the MSD received by the PSAP test point does not deviate from the exact recorded time of the trigger activation by more than 60 seconds.

▼**M1**

3.2. If the automatic test eCall could not be performed successfully due to vehicle-external factors, it shall be permissible to verify the automatic trigger following the impact via the internal record transaction function of the in-vehicle system. This register shall be capable to store received trigger signals in non-volatile memory. The test engineer shall have access to the data stored in the in-vehicle system and shall verify that no record of automatic trigger signal is stored before the impact event and that a record of an automatic trigger signal is stored after the impact event.

3.3. If the test eCall was performed with the vehicle connected to an off-vehicle power supply (in cases where the impact test was carried out with the standard vehicle power supply not installed), verify that the on-board electrical system feeding the eCall in-vehicle system remained intact. This shall be verified by a record of a test engineer confirming a successful check of the integrity of the on-board electrical system including the dummy in-vehicle power source (visual inspection for mechanical damage to either the power source's mounting bracket or its structure) and the connections via its terminals.

3.4. Verification of Power-Supply test

3.4.1. The requirement is determined to have been passed if the eCall in-vehicle system is capable to communicate for the required period, as specified in point 2 of Annex X. Otherwise, the test is determined to have been failed.

▼B

ANNEX III

Crash resistance of audio equipment

1. Requirements

1.1. Performance requirements

1.1.1. The assessment of the crash resistance of the eCall audio equipment of vehicles with eCall in-vehicle systems installed, carried out in accordance with point 2, shall be considered satisfactory if the following requirements are demonstrated post-impact as regards the frontal impact test as well as the side impact test, as applicable.

▼M1

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▼B

1.1.3. Voice communication: The eCall system shall allow hands-free voice communication (send and receive direction) of sufficient intelligibility between vehicle occupants and an operator.

2. Test procedure

▼M1

2.1. Purpose of the audio equipment crash resistance test procedure

The purpose of this test is to verify that loudspeaker(s) and microphone(s) are successfully connected and that the audio equipment remained functional after the vehicle has been subjected to the frontal impact or the side impact test.

▼B

2.2. The following verification test shall be performed on a vehicle with the eCall in-vehicle system installed that has been subjected to a full-scale impact according to Regulation No 94, Annex 3 for frontal impact or UN Regulation No 95, Annex 4 for side impact, as set out in point 1.1.1 above.

2.3. Overview of test procedure

▼M1

2.3.1. The sustained functionality of the audio equipment shall be verified by performing a test eCall after the impact test and using the voice communication channel between the vehicle and the PSAP test point.

▼B

2.3.2. Two test engineers, positioned in the vehicle (near-end tester) and at the PSAP test point (far-end tester) respectively, successively transmit (read and listen) pre-defined, phonetically balanced sentences in single talk mode.

2.3.3. The testers are required to assess whether they were able to understand the meaning of the transmission in the send and receive directions.

2.4. Arrangement of testers

2.4.1. The test shall be performed in a quiet environment, with a background noise level of not more than 50 dB(A) and that is free from any noise sources that might otherwise disrupt the tests.

▼B

2.4.2. The near-end tester shall be positioned so that his head is close to a normal seating position on the driver's seat of the impacted vehicle. The tester shall use the in-vehicle audio equipment in the original arrangement.

2.4.3. The far-end tester shall be positioned away from the vehicle with sufficient separation so that speech in normal loudness from one tester cannot be understood without any aids by the other tester.

▼M1

2.5. Test conditions

2.5.1. Before performing the test eCall, ensure that:

(a) one of the connection procedures specified in point 2.7 of Annex I, as agreed between technical service and manufacturer, will be applied for any test eCall;

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based eCall in-vehicle system or the eCall in-vehicle system is registered so that a call can be started from the PSAP test point;

(c) a false eCall to a genuine PSAP cannot be made over the live network;

(d) if applicable, the TPS system is deactivated or will automatically switch to the 112-based system;

(e) the vehicle ignition or master control switch is activated;

(f) an external power supply source can be used, in case where power capacity (main and back-up) is no longer available following the test procedure set out in Annex II.

▼B

2.5.2. Where it is possible to adapt the volume setting, the maximum volume control setting in send and receive direction at the near-end and at the far-end shall be chosen. The volume control settings at the far-end may be decreased during the test if required for better intelligibility.

2.5.3. If possible, no mobile networks that have an influence on the hands-free performance (e.g. echo, AGC, noise reduction, etc.) should be chosen for the connection. For simulated networks, if possible, DTX shall be switched off, the full rate codec shall be used (for GSM standard) and the highest bit rate of 12,2 kbit/s shall be used (for AMR codecs).

▼M1

2.6. Test method

2.6.1. Perform a test eCall by applying a manual trigger via the in-vehicle HMI and wait until the loudspeaker(s) and microphone(s) are connected and ready for voice communication or perform a call from the PSAP test point to the eCall in-vehicle system.

▼B

2.6.2. Exchange of test messages

2.6.2.1. Receive direction

2.6.2.1.1. The far-end tester shall select and read one sentence pair of the list provided in the Appendix. The tester shall read the sentences in a normal volume as used in phone calls.

▼B

2.6.2.1.2. The near-end tester shall assess whether the voice transmission in the receive direction was intelligible: The test in receive direction is passed if the near-end tester, resting in his original seating position, was able, with any feasible effort, to understand the full meaning of the transmission.

2.6.2.1.3. If required for the assessment, the near-end tester can request from the far-end tester to transmit additional sentence pairs.

2.6.2.2. Send direction

2.6.2.2.1. The near-end tester shall select and, resting in his original seating position, read one sentence pair of the list provided in the Appendix. The tester shall read the sentences in a normal volume as used in phone calls.

2.6.2.2.2. The far-end tester shall assess whether the voice transmission in the send direction was intelligible: The test in send direction is passed if the far-end tester was able, with any feasible effort, to understand the full meaning of the transmission.

2.6.2.2.3. If required for the assessment, the far-end tester can request from the near-end tester to transmit additional sentence pairs.

▼M1

2.6.3. Clear down the test eCall using the appropriate PSAP test point command (e.g. hang up).

2.6.4. If the requirements cannot be fulfilled due to impairments introduced by the PSAP test point or the transmission medium, the test eCall may be repeated, if required in an adapted test setup.

▼B

2.7. Connection procedures

2.7.1. The connection procedures defined in point 2.7 of Annex I to this Regulation shall apply.

▼B

Appendix

Test sentences

1. The following test sentence pairs, as defined in ITU-T P.501, Annex B, shall be used for the exchange of test messages in the send and receive directions.

2. Test sentence pairs in the language most commonly spoken by the testers shall be selected from the list below. If the testers are not familiar with any of the languages, alternative sentences in a familiar language, preferably phonetically balanced, shall be used.

3. Test sentence pairs

3.1. Dutch

(a) Dit product kent nauwelijks concurrentie.

Hij kende zijn grens niet.

(b) Ik zal iets over mijn carrière vertellen.

Zijn auto was alweer kapot.

▼M1

(c) Zij kunnen de besluiten nemen.

De meeste mensen hadden het wel door.

▼B

(d) Ik zou liever gaan lopen.

Willem gaat telkens naar buiten.

3.2. English

(a) These days a chicken leg is a rare dish.

The hogs were fed with chopped corn and garbage.

(b) Rice is often served in round bowls.

A large size in stockings is hard to sell.

(c) The juice of lemons makes fine punch.

Four hours of steady work faced us.

(d) The birch canoe slid on smooth planks.

Glue the sheet to the dark blue background.

3.3. Finnish

(a) Ole ääneti tai sano sellaista, joka on parempaa kuin vaikeneminen.

Suuret sydämet ovat kuin valtameret, ne eivät koskaan jäädy.

(b) Jos olet vasara, lyö kovaa. Jos olet naula, pidä pääsi pystyssä.

Onni tulee eläen, ei ostaen.

(c) Rakkaus ei omista mitään, eikä kukaan voi sitä omistaa.

Naisen mieli on puhtaampi, hän vaihtaa sitä useammin.

(d) Sydämellä on syynsä, joita järki ei tunne.

On opittava kärsimään voidakseen elää.

▼B

3.4. French

(a) On entend les gazouillis d'un oiseau dans le jardin.

La barque du pêcheur a été emportée par une tempête.

(b) Le client s'attend à ce que vous fassiez une réduction.

Chaque fois que je me lève ma plaie me tire.

(c) Vous avez du plaisir à jouer avec ceux qui ont un bon caractère.

Le chevrier a corné pour rassembler ses moutons.

(d) Ma mère et moi faisons de courtes promenades.

La poupée fait la joie de cette très jeune fille.

3.5. German

(a) Zarter Blumenduft erfüllt den Saal.

Wisch den Tisch doch später ab.

(b) Sekunden entscheiden über Leben.

Flieder lockt nicht nur die Bienen.

(c) Gegen Dummheit ist kein Kraut gewachsen.

Alles wurde wieder abgesagt.

(d) Überquere die Strasse vorsichtig.

Die drei Männer sind begeistert.

3.6. Italian

(a) Non bisogna credere che sia vero tutto quello che dice la gente. Tu non conosci ancora gli uomini, non conosci il mondo.

Dopo tanto tempo non ricordo più dove ho messo quella bella foto, ma se aspetti un po' la cerco e te la prendo.

(b) Questo tormento durerà ancora qualche ora. Forse un giorno poi tutto finirà e tu potrai tornare a casa nella tua terra.

Lucio era certo che sarebbe diventato una persona importante, un uomo politico o magari un ministro. Aveva a cuore il bene della società.

(c) Non bisogna credere che sia vero tutto quello che dice la gente tu non conosci ancora gli uomini, non conosci il mondo.

Dopo tanto tempo non ricordo più dove ho messo quella bella foto ma se aspetti un po' la cerco e te la prendo.

(d) Questo tormento durerà ancora qualche ora. Forse un giorno poi tutto finirà e tu potrai tornare a casa nella tua terra.

Lucio era certo che sarebbe diventato una persona importante, un uomo politico o magari un ministro, aveva a cuore il bene della società.

3.7. Polish

(a) Pielęgniarki były cierpliwe.

Przebiegał szybko przez ulicę.

(b) Ona była jego sekretarką od lat.

Dzieci często płaczą kiedy są głodne.

(c) On był czarującą osobą.

Lato wreszcie nadeszło.

(d) Większość dróg było niezmiernie zatłoczonych.

Mamy bardzo entuzjastyczny zespół.

▼B

3.8. Spanish

(a) No arroje basura a la calle.

Ellos quieren dos manzanas rojas.

(b) No cocinaban tan bien.

Mi afeitadora afeita al ras.

(c) Ve y siéntate en la cama.

El libro trata sobre trampas.

(d) El trapeador se puso amarillo.

El fuego consumió el papel.

▼B

ANNEX IV

Co-existence of third party services (TPS) with the 112-based eCall in-vehicle systems

1. Requirements

1.1. The following requirements apply to 112-based eCall in-vehicle systems, STUs and (optionally for) components that shall be used in conjunction with a TPS eCall in-vehicle system.

▼M1

1.1.1. In case of an optional TPS eCall system, the TPS eCall related tests set out in section 9.8 of standard EN 16454:2023 remain applicable for the Declaration of Conformity of eCall in-vehicle system after 31 December 2025.

▼B

1.2. Performance requirements

1.2.1. The 112-based system shall be deactivated as long as the TPS system is active and does function.

1.2.2. The 112-based system shall be automatically triggered in the event that the TPS system is triggered but does not function.

1.3. Documentation requirements

1.3.1. The manufacturer shall provide the technical service with an explanation of the design provisions built into the TPS system to ensure automatic triggering of the 112-based system ('fall-back procedure') in the event that the TPS system does not function. This documentation shall describe the principles of the changeover mechanism.

1.3.2. The documentation shall be supported by an analysis which shows, in overall terms, any hardware or software failure conditions that would result in an inability of the TPS system to perform a successful call and how the TPS system will behave on the occurrence of these.

This may be based on a Failure Mode and Effect Analysis (FMEA), a Fault Tree Analysis (FTA) or any appropriate similar process as agreed between the technical service and the manufacturer.

The chosen analytical approach(es) shall be established and maintained by the manufacturer and shall be made open for inspection by the technical service at the time of the type-approval.

2. Test procedure

2.1. Purpose of the TPS co-existence test procedure

The purpose of this test procedure is to verify for eCall in-vehicle systems that shall be used in conjunction with a TPS eCall in-vehicle system, that there is only one system active at a time and that the 112-based system is triggered automatically in the event that the TPS system does not function.

2.2. The following tests shall be performed either on a vehicle with an eCall in-vehicle system installed or on a representative arrangement of parts.

▼**M1**

2.3. The deactivation of the 112-based system while the TPS system is active, shall be verified by performing a manually triggered test eCall.

2.3.1. Before performing the test eCall, ensure:

(a) that one of the connection procedures specified in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

(b) that the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) that the TPSP test point is available to receive an eCall emitted by the TPS system;

(d) that a false eCall to a genuine PSAP cannot be made over the live network; and

(e) that the vehicle ignition or master control switch is activated.

2.3.2. Perform a test eCall by applying a manual trigger of the TPS system.

2.3.3. Verify that:

(a) an eCall was established with the TPSP test point by a successful voice connection to the TPSP test point;

(b) data was received properly at the TPSP test point including at least the MSD; and

(c) no eCall was attempted or established with the PSAP test point by a record of the PSAP test point showing that it did not receive an eCall.

2.3.4. Clear down the eCall using the appropriate TPSP test point command (e.g. hang up).

▼**B**

2.3.5. If the call attempt of the TPS system fails during the test, the test procedure may be repeated.

▼**M1**

2.4. The fall-back procedure and activation of the 112-based system, in the event that the TPS system does not function, shall be verified by performing a manually triggered test eCall.

2.4.1. Modify the TPS system to simulate a failure, selected at the discretion of the technical service that shall result in a fall-back procedure based on the documentation provided by the manufacturer. Such selection shall be documented in the test report.

2.4.2. Before performing the test eCall, ensure that:

(a) one of the connection procedures defined in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

▼**M1**

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) a false eCall to a genuine PSAP cannot be made over the live network; and

(d) the vehicle ignition or master control switch is activated.

2.4.3. Perform a test eCall by applying a manual trigger of the TPS system.

2.4.4. Verify that:

(a) an eCall was established with the PSAP test point by a successful voice connection to the PSAP test point; and

(b) an MSD was received by the PSAP test point. That shall be verified by a record of the PSAP test point showing that an MSD emitted from the eCall system following the trigger was received and successfully decoded.

2.4.5. Clear down the test eCall using the appropriate PSAP test point command (e.g. hang up).

▼B

ANNEX V

Automatic triggering mechanism

1. Requirements

1.1. The following requirements apply to vehicles with eCall in-vehicle systems installed.

1.2. Documentation requirements

1.2.1. The manufacturer shall provide a statement which affirms that the strategy chosen to trigger an automatic eCall ensures triggering also in accident configurations dissimilar from and/or of a lower severity than the collisions simulated in the applicable full-scale crash tests in UN Regulation No 94 and UN Regulation No 95.

1.2.2. The manufacturer shall choose the collision typology and severity and will demonstrate that it is significantly different than the full-scale crash tests.

1.2.3. The manufacturer shall provide the type-approval authority with an explanation and technical documentation which shows, in overall terms, how this is achieved.

1.2.3.1. Documentation that shows, to the satisfaction of the type-approval authority, that the activation of supplemental restraint systems and the severity level, chosen at the discretion of the manufacturer, also induces an automatic eCall shall be considered satisfactory.

1.2.3.2. Documentation that shows, to the satisfaction of the type-approval authority, the strategy to prevent unjustified eCalls from being made in case of impacts of a severity level that is not considered a severe accident. Moreover, failure mode analysis shall be provided which shows that any hardware or software faults shall not result in automatic triggering of an eCall.

1.2.3.3. Airbag control unit specification drawings, specification data notes, sensitivity drawings, relevant circuit diagrams or similar documents considered equivalent by the type-approval authority would be suitable means to demonstrate this connection.

1.2.3.4. The extended documentation package shall remain strictly confidential. It may be kept by the approval authority, or, at the discretion of the approval authority, may be retained by the manufacturer. In case the manufacturer retains the documentation package, that package shall be identified and dated by the approval authority once reviewed and approved. It shall be made available for inspection by the approval authority at the time of approval or at any time during the validity of the approval.

▼B

ANNEX VI

Technical requirements for compatibility of eCall in-vehicle systems with the positioning services provided by the Galileo and the EGNOS systems

1. Requirements

1.1. Compatibility requirements

1.1.1. The 'Galileo system compatibility' shall be: the reception and processing of the signals from the Open Service of Galileo, using it in the computation of the final position.

1.1.2. The 'EGNOS system compatibility' shall be: the reception of the corrections from the Open Service of EGNOS and its application to the GNSS signals, in particular GPS.

1.1.3. The compatibility of the eCall in-vehicle systems with the positioning services provided by the Galileo and the EGNOS systems shall be compliant with respect to positioning capabilities in section 1.2 and demonstrated by performing the test methods in section 2.

1.1.4. The testing procedures in section 2.2 can be performed either on the eCall unit including post processing ability or directly on the GNSS receiver being a part of the eCall.

1.2. Performance requirements

1.2.1. The GNSS receiver shall be able to output the navigation solution in a NMEA-0183 protocol format (RMC, GGA, VTG, GSA and GSV message). The eCall setup for NMEA-0183 messages output shall be described in the operation manual.

1.2.2. The GNSS receiver being a part of the eCall shall be capable of receiving and processing individual GNSS signals in L1/E1 band from at least two global navigation satellite systems, including Galileo and GPS.

1.2.3. The GNSS receiver being a part of the eCall shall be capable of receiving and processing combined GNSS signals in L1/E1 band from at least two global navigation satellite systems, including Galileo and GPS; and SBAS.

1.2.4. The GNSS receiver being a part of the eCall shall be able to provide positioning information in WGS-84 coordinate system.

1.2.5. Horizontal position error shall not exceed:

— under open sky conditions: 15 metres at confidence level 0,95 probability with Position Dilution of Precision (PDOP) in the range from 2,0 to 2,5;

— in urban canyon conditions: 40 metres at confidence level 0,95 probability with Position Dilution of Precision (PDOP) in the range from 3,5 to 4,0.

1.2.6. The specified requirements for accuracy shall be provided:

— at speed range from 0 to [140] km/h;

— linear acceleration range from 0 to [2] G.

▼B

1.2.7. Cold start time to first fix shall not exceed

— 60 seconds for signal level down to minus 130 dBm;

— 300 seconds for signal level down to minus 140 dBm.

1.2.8. GNSS signal re-acquisition time after block out of 60 seconds at signal level down to minus 130 dBm shall not exceed 20 seconds after recovery of the navigation satellite visibility.

1.2.9. Sensitivity at receiver input shall be:

— GNSS signals detection (cold start) do not exceed 3 600 seconds at signal level on the antenna input of the eCall of minus 144 dBm;

— GNSS signals tracking and navigation solution calculation is available for at least 600 seconds at signal level on the antenna input of the eCall of minus 155 dBm;

— Re-acquisition of GNSS signals and calculation of the navigation solution is possible and does not exceed 60 seconds at signal level on the antenna input of the eCall of minus 150 dBm.

1.2.10. The GNSS receiver shall be able to obtain a position fix at least every second.

2. Test methods

2.1. Test conditions

2.1.1. The test object is the eCall, which includes a GNSS receiver and a GNSS antenna, specifying navigation characteristics and features of the tested system.

2.1.2. The number of the eCall test samples shall be at least 3 pieces and can be tested in parallel.

2.1.3. The eCall is provided for the test with the installed SIM-card, operation manual and the software (provided on electronic media).

2.1.4. The attached documents shall contain the following data:

— device serial number;

— hardware version;

— software version;

— device provider identification number;

— relevant technical documentation to perform the tests.

2.1.5. Tests are carried out in normal climatic conditions in accordance with standard ISO 16750-1:2006:

— air temperature 23 (± 5) ° C;

— relative air humidity of 25 % to 75 %.

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2.1.6. Tests of the eCall in respect of its GNSS receiver shall be performed with the test and auxiliary equipment specified in Table 1.

Table 1

Recommended list of measurement instruments, test and auxiliary equipment

Equipment name	Required technical characteristics of test equipment	
	Scale range	Scale accuracy
Global navigation satellite system simulator of Galileo and GPS signals	Number of simulated signals: at least 12	Mean square deviation of random accuracy component of pseudo-range to Galileo and GPS satellites not more than: — stadiometric code phase: 0,1 metres; — communication carrier phase: 0,001 metres; — pseudovelocity: 0,005 metres/second.
Digital stopwatch	Maximum count volume: 9 hours 59 minutes 59,99 seconds	Daily variation at 25 (± 5) °C not more than 1,0 seconds. Time discreteness 0,01 seconds.
Vector network analyser	Frequency range: 300 kHz .. 4 000 kHz Dynamic range: (minus 85 .. 40) dB	Accuracy F = ± $1 \cdot 10^{-6}$ kHz Accuracy D = (0,1 .. 0,5) dB
Low-noise amplifier	Frequency range: 1 200 .. 1 700 MHz Noise coefficient: not more 2,0 dB Amplifier gain coefficient: 24 dB	
Attenuator 1	Dynamic range: (0 .. 11) dB	Accuracy ± 0,5 dB
Attenuator 2	Dynamic range: (0 .. 110) dB	Accuracy ± 0,5 dB
Power source	Range of direct current voltage setting: from 0,1 to 30 volts	Accuracy V = ± 3 %
	Current intensity of output voltage: at least 3 amperes	Accuracy A = ± 1 %

Note: it is allowed to apply other similar types of equipment providing determination of characteristics with the required accuracy.

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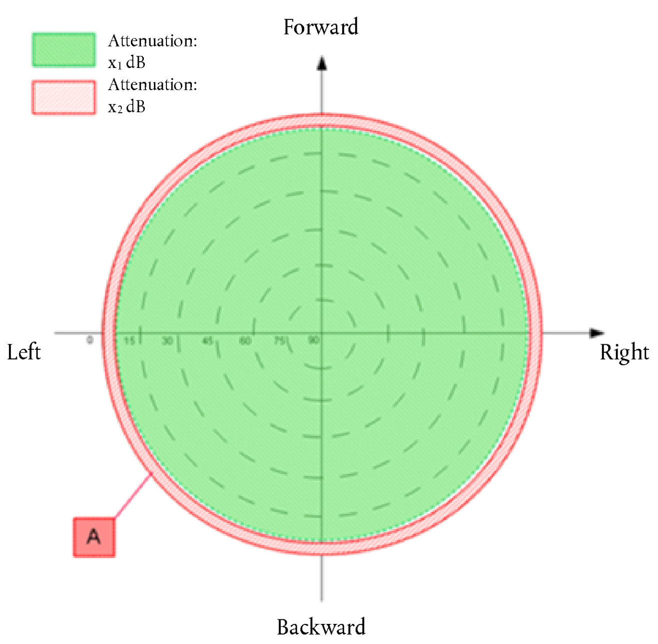
2.1.7. Unless otherwise specified, GNSS signal simulation shall follow 'Open sky' pattern as shown in Figure 1.

Figure 1

Open sky definition

Zone	Elevation range (degrees)	Azimuth range (degrees)
A	0 – 5	0 – 360
Background	Area out of Zone A	



2.1.8. Open Sky plot — Attenuation:

	0 dB
A	– 100 dB or signal is switched off

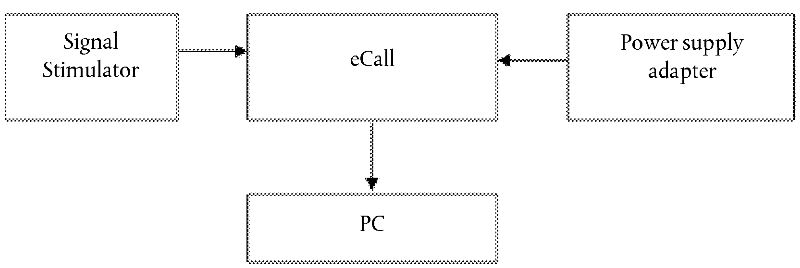
2.2. Test procedures

2.2.1. NMEA-0183 messages output test.

2.2.1.1. Make connections according to Figure 2.

Figure 2

Diagram of test stand



▼B

2.2.1.2. Prepare and turn on the eCall. By means of operation manual and developer software, set up the GNSS receiver for receiving signals from Galileo, GPS and SBAS. Set up the GNSS receiver to output NMEA-0183 messages (messages RMC, GGA, VTG, GSA and GSV).

2.2.1.3. Set up the simulator according to the simulator user guide. Initialize simulator script with the parameters, given in Table 2 for Galileo, GPS and SBAS signals.

Table 2

Main parameters of simulation script for static scenario

Simulated parameter	Value
Test duration, hh:mm:ss	01:00:00
Output frequency	1 hertz
eCall location	Any specified land point between latitude range 80°N and 80°S in coordinate system WGS-84
Troposphere:	Standard predefined model by the GNSS simulator
Ionosphere:	Standard predefined model by the GNSS simulator
PDOP value in the test interval	2,0 ≤ PDOP ≤ 2,5
Simulated signals	— Galileo (E1 frequency band OS); — GPS (L1 frequency band C/A code); — combined Galileo/GPS/SBAS.
Signal strength: — GNSS Galileo; — GNSS GPS.	 minus 135 dBm; minus 138,5 dBm.
Number of simulated satellites:	— at least 6 Galileo satellites; — at least 6 GPS satellites; — at least 2 SBAS satellites

2.2.1.4. By means of corresponding serial interface, set the connection between the eCall and PC. Control the possibility of receiving navigation information via NMEA-0183 protocol. The value of field 6 in the GGA messages is set to ‘2’.

2.2.1.5. Test results are considered successful if navigation information via NMEA-0183 protocol is received in all the eCall samples.

2.2.1.6. The test of NMEA-0183 messages output and the assessment of the positioning accuracy in autonomous static mode can be combined.

▼B

2.2.2. Assessment of positioning accuracy in autonomous static mode.

2.2.2.1. Make connections according to Figure 2.

2.2.2.2. Prepare and turn on the eCall. By means of developer software, make sure that the GNSS receiver is set up for receiving Galileo, GPS and SBAS combined signals. Set up the GNSS receiver to output messages according to the NMEA-0183 protocol (GGA, RMC, VTG, GSA and GSV messages).

2.2.2.3. Set up the simulator in accordance with its operational manual. Start simulation of combined Galileo, GPS and SBAS signals script with the set parameters given in Table 2.

2.2.2.4. Set up the recording of NMEA-0183 messages after receiving the navigation solution. Up to the moment the simulation script is complete, the NMEA-0183 messages are output by the GNSS receiver to a file.

2.2.2.5. Upon receiving the navigation solution set up recording of NMEA-0183 messages output by the GNSS receiver to a file, up to the moment the simulation script is complete.

2.2.2.6. Extract coordinates: latitude (B) and longitude (L) contained in GGA (RMC) messages.

2.2.2.7. Calculate the systematic inaccuracy of coordinate's determination on stationary intervals according to formulas (1), (2), for example for latitude coordinate (B):

$$(1) \qquad \Delta B(j) = B(j) - B_{truej},$$

$$(2) \qquad dB = \frac{1}{N} \cdot \sum_{j=1}^{N} \Delta B(j),$$

— B_{truej} is the actual value of B coordinate in j time moment, in arc-seconds.

— B(j) is the determined value of B coordinate in j time moment by the GNSS receiver, in arc-seconds.

— N is the amount of GGA (RMC) messages, received during the test of GNSS receiver.

2.2.2.8. Similarly calculate the systematic inaccuracy of L (longitude) coordinate.

2.2.2.9. Calculate standard deviation (SD) value according to formula (3) for B coordinate:

$$(3) \qquad \sigma_B = \sqrt{\frac{\sum_{j=1}^{N} (\Delta B(j) - dB)^2}{N - 1}},$$

2.2.2.10. Similarly calculate the SD value for L (longitude) coordinate.

2.2.2.11. Convert calculated coordinates and SD values of latitude and longitude determination from arc-seconds to meters according to formulas (4) – (5).

▼**B**

2.2.2.12. For latitude:

$$(4\text{-}1) \qquad dB(\text{м}) = 2 \cdot \frac{a \cdot (1 - e^2)}{(1 - e^2 \sin^2\varphi)^{3/2}} \cdot \frac{0{,}5'' \cdot \pi}{180 \cdot 3\,600''} \cdot dB,$$

$$(4\text{-}2) \qquad \sigma_B(\text{м}) = 2 \cdot \frac{a \cdot (1 - e^2)}{(1 - e^2 \sin^2\varphi)^{3/2}} \cdot \frac{0{,}5'' \cdot \pi}{180 \cdot 3\,600''} \cdot \sigma_B,$$

2.2.2.13. For longitude:

$$(5\text{-}1) \qquad dL(\text{м}) = 2 \cdot \frac{a \cdot \cos\varphi}{\sqrt{1 - e^2 \sin^2\varphi}} \cdot \frac{0{,}5'' \cdot \pi}{180 \cdot 3\,600''} \cdot dL,$$

$$(5\text{-}2) \qquad \sigma_L(\text{м}) = 2 \cdot \frac{a \cdot \cos\varphi}{\sqrt{1 - e^2 \sin^2\varphi}} \cdot \frac{0{,}5'' \cdot \pi}{180 \cdot 3\,600''} \cdot \sigma_L,$$

— *a* — Semi-major axis of ellipsoid, metres

— *e* — first eccentricity, [0 – 1]

— *φ* — determined value of latitude, radians.

2.2.2.14. Calculate horizontal position error according to formula (6):

$$(6) \qquad \Pi = \sqrt{dB^2(m) + dL^2(m)} + 2 \cdot \sqrt{\sigma_B^2(m) + \sigma_L^2(m)},$$

2.2.2.15. Repeat test procedures according to 2.2.2.3 – 2.2.2.14 for GNSS Galileo signals with simulation parameters, given in Table 2.

2.2.2.16. Repeat test procedures according to 2.2.2.3 – 2.2.2.14 only for GPS GNSS signals with simulation parameters, given in Table 2.

2.2.2.17. Repeat test procedures according to 2.2.2.3 – 2.2.2.16 with other eCall samples, provided for the test.

2.2.2.18. Determine average values according to (6) obtained for all tested eCall samples.

2.2.2.19. Tests results are considered satisfactory if horizontal position errors as defined by formula (6) obtained with all eCall samples do not exceed 15 metres under open sky conditions at confidence level 0,95 probability for all simulation scripts.

2.2.3. Assessment of positioning accuracy in autonomous dynamic mode.

2.2.3.1. Repeat test procedures described in section 2.2.2, but 2.2.2.15 – 2.2.2.16 with simulation script for manoeuvring movement, given in Table 3.

▼B

Table 3

Main parameters of simulation script for manoeuvring movement

Simulated parameter	Value
Test duration, hh:mm:ss	01:00:00
Output frequency	1 hertz
eCall location	Any specified land point between latitude range 80°N and 80°S in coordinate system WGS-84
Model of movement: — speed, km/h; — turning radius, metres; — turning acceleration, metres/second^2.	Manoeuvring movement 140 500 0,2
Troposphere: Ionosphere:	Standard predefined model by the GNSS simulator Standard predefined model by the GNSS simulator
PDOP value in the test time interval	2,0 ≤ PDOP ≤ 2,5
Simulated signals	Combined Galileo/GPS/SBAS
Signal strength: — GNSS Galileo; — GNSS GPS.	 minus 135 dBm; minus 138,5 dBm.
Number of simulated satellites:	— at least 6 Galileo satellites; — at least 6 GPS satellites; — at least 2 SBAS satellites

2.2.3.2. Determine average values according to (6) obtained for all tested eCall samples.

2.2.3.3. Tests results are considered satisfactory if horizontal position errors obtained with all eCall samples do not exceed 15 metres under open sky conditions at confidence level 0,95 probability.

2.2.4. Movement in shadow areas, areas of intermittent reception of navigation signals and urban canyons.

▼B

2.2.4.1. Repeat test procedures described in section 2.2.3 for simulation script for movement in shadow areas and areas of intermittent reception of navigation signals (given in Table 4) with an urban canyon signal pattern described in Figure 3.

Table 4

Main parameters of movement in shadow areas and areas of intermittent reception of navigation signals

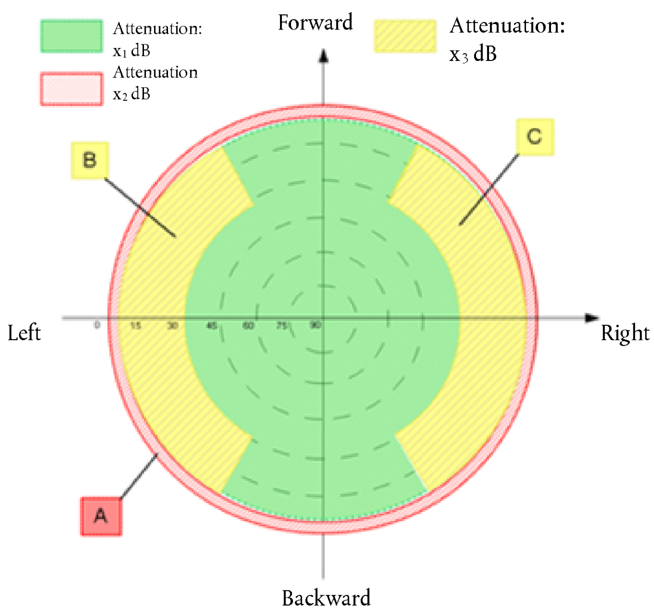
Simulated parameter	Value
Test duration, hh:mm:ss	01:00:00
Output frequency	1 hertz
eCall location	Any specified land point between latitude range 80°N and 80°S in coordinate system WGS-84
Model of movement:	Manoeuvring movement
— speed, km/h;	140
— turning radius, metres;	500
— turning acceleration, metres/second^2.	0,2
Satellite visibility:	
— signal visibility intervals, seconds;	300
— signal absence intervals, seconds.	600
Troposphere:	Standard predefined model by the GNSS simulator
Ionosphere:	Standard predefined model by the GNSS simulator
PDOP value in the test time interval	$3,5 \leq PDOP \leq 4,0$
Simulated signals	Combined Galileo/GPS/SBAS
Signal strength:	
— GNSS Galileo;	minus 135 dBm;
— GNSS GPS.	minus 138,5 dBm.
Number of simulated satellites:	— at least 6 Galileo satellites; — at least 6 GPS satellites; — at least 2 SBAS satellites

▼B

Figure 3

Urban canyon definition

Zone	Elevation range (degrees)	Azimuth range (degrees)
A	0 – 5	0 – 360
B	5 – 30	210 – 330
C	5 – 30	30 – 150
Background	Area out of Zone A, B, C	



2.2.4.2. Urban canyon plot- Attenuation:

	0 dB
B	– 40 dB
C	– 40 dB
A	– 100 dB or signal is switched off

2.2.4.3. Tests results are considered satisfactory if horizontal position errors obtained with all eCall samples do not exceed 40 metres in urban canyon conditions at confidence level 0,95 probability.

2.2.5. Cold start time to first fix test.

2.2.5.1. Prepare and turn on the eCall. By means of developer software, make sure that GNSS module is set to receive Galileo and GPS signals.

2.2.5.2. Delete all position, velocity, time, almanac and ephemeris data from the GNSS receiver.

▼B

2.2.5.3. Set up the simulator according to the simulator user guide. Initialize simulator script with the parameters, given in Table 2 for Galileo and GPS signals with signal level minus 130 dBm.

2.2.5.4. By means of a stopwatch, measure time interval between signal simulation start and the first navigation solution result.

2.2.5.5. Conduct test procedures according to 2.2.5.2 – 2.2.5.4 at least 10 times.

2.2.5.6. Calculate average time to first fix in cold start mode based on measurements for all eCall samples, provided for the test.

2.2.5.7. The test result is considered to be positive, if average values of time to first fix calculated as described in 2.2.5.6, do not exceed 60 seconds for signal level down to minus 130 dBm for all the simulated signals.

2.2.5.8. Repeat test procedure according to 2.2.5.1 – 2.2.5.5 with signal level minus 140 dBm.

2.2.5.9. The test result according to 2.2.5.8 is considered to be positive, if average values of time to first fix, calculated as described in 2.2.5.6 do not exceed 300 seconds for signal level down to minus 140 dBm for all the simulated signals.

2.2.6. Test of re-acquisition time of tracking signals after block out of 60 seconds.

2.2.6.1. Prepare and turn on the eCall according to operational manual. By means of the developer software, make sure that GNSS receiver is set up to receive Galileo and GPS signals.

2.2.6.2. Set up the simulator according to the simulator user guide. Initialize simulator script with the parameters, given in Table 2 for Galileo and GPS signals with signal level minus 130 dBm.

2.2.6.3. Wait for 15 minutes and make sure the GNSS receiver has calculated eCall position.

2.2.6.4. Disconnect the GNSS antenna cable from the eCall and connect it again after time interval of 60 seconds. By means of stopwatch, determine time interval between cable connection moment and restoration of satellites tracking and calculation of the navigation solution.

2.2.6.5. Repeat test procedure according to 2.2.6.4 at least 10 times.

2.2.6.6. Calculate average value of re-acquisition time of satellite tracking signals by the eCall for all performed measurements and all eCall samples provided for the test.

2.2.6.7. The test result is considered to be positive, if average values of re-acquisition time after block out of 60 seconds measured as described in 2.2.6.6, do not exceed 20 seconds.

▼B

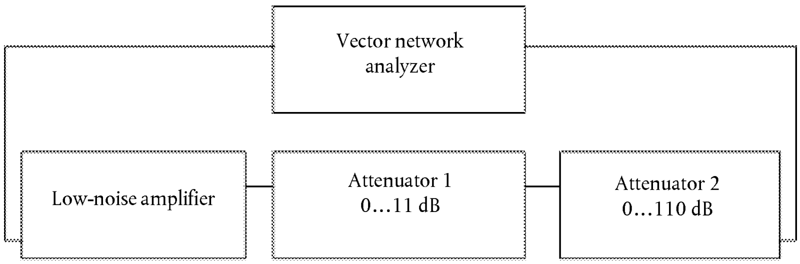
2.2.7. Test of GNSS receiver sensitivity in cold start mode, tracking mode, and re-acquisition scenario.

2.2.7.1. Turn on the vector network analyser. Calibrate the vector network analyser according to its operational manual.

2.2.7.2. Set up the diagram according to Figure 4.

Figure 4

Diagram of path calibration

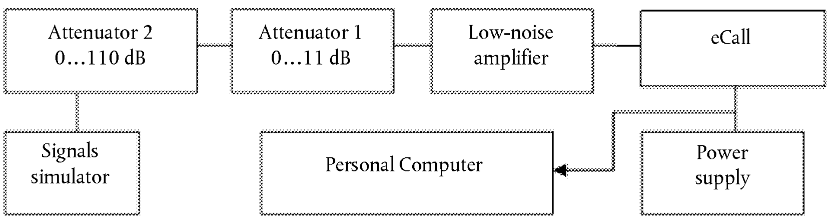


2.2.7.3. Set zero signal path attenuation on attenuators. Measure the frequency response for a given signal path in the E1/L1 band of Galileo/GPS, respectively. Record the average path transmission factor in [dB] in this frequency band.

2.2.7.4. Assemble the circuit shown in Figure 5.

Figure 5

Arrangement for evaluation of GNSS module sensitivity



2.2.7.5. Prepare and turn on eCall according to operational manual. By means of developer software make sure that GNSS receiver is set to receive Galileo and GPS signals. Clear the GNSS receiver RAM such that the ‘cold’ start mode of the GNSS receiver of the eCall is achieved. Check that the position, velocity and time information is reset.

2.2.7.6. Prepare GNSS signals simulator according to its operation manual. Start Galileo and GPS signals simulation script, with parameters given in Table 2. Set output power level of the simulator to minus 144 dBm.

2.2.7.7. By means of a stopwatch, measure time interval between signal simulation start and the first navigation solution result.

2.2.7.8. Set the signal path attenuation on attenuators such that the signal on eCall antenna input is equal to minus 155 dBm.

▼B

2.2.7.9. By means of a stopwatch, verify that the eCall still provides navigation solution for at least 600 seconds.

2.2.7.10. Set the signal path attenuation on attenuators such that the signal on eCall antenna input is equal to minus 150 dBm.

2.2.7.11. Disconnect the GNSS antenna cable from the eCall and connect it again after time interval of 20 seconds.

2.2.7.12. By means of stopwatch, determine time interval between cable connection moment and restoration of satellites tracking and calculation of the navigation solution.

2.2.7.13. The test result is considered to be positive in case:

— the value of time to first fix in 'cold' start mode, as measured in 2.2.7.7, do not exceed 3 600 seconds at signal level on the antenna input of the eCall of minus 144 dBm in all the eCall samples;

— the GNSS navigation solution is available for at least 600 seconds at signal level on the antenna input of the eCall of minus 155 dBm as measured in 2.2.7.9 in all the eCall samples;

— and re-acquisition of GNSS signals and calculation of the navigation solution at signal level on the antenna input of the eCall of minus 150 dBm is possible and time interval measured in 2.2.7.12 does not exceed 60 seconds in all the eCall samples.

▼B

ANNEX VII

In-vehicle system self-test

1. Requirements

1.1. The following requirements apply to vehicles with eCall in-vehicle system installed, STUs and (optionally for) components.

1.2. Performance requirements

1.2.1. The eCall system shall carry out a self-test at each system power-up.

1.2.2. The self-test function shall monitor at least the technical items listed in the Table.

1.2.3. A warning in form of either a visual tell-tale or a warning message in a common space shall be provided in case a failure is detected by the self-test function.

1.2.3.1. It shall remain activated while the failure is present.

1.2.3.2. It may be cancelled temporarily, but shall be repeated whenever the ignition or vehicle master control switch is being activated.

1.3. Documentation requirements

▼M1

1.3.1. The manufacturer shall provide the technical service and the type-approval authority with documentation in accordance with the Table, which shall contain for each item the technical principle applied to monitor the item.

▼B

Table

Template of information for self-test function

Item	Technical principle applied for monitoring
eCall ECU is in working order (e.g. no internal hardware failure, processor/ memory is ready, logic function in expected default state)	
►M1 Mobile network antenna is connected ◄	
Mobile network communication device is in working order (no internal hardware failure, responsive)	
►M1 GNSS antenna is connected ◄	
GNSS receiver is in working order (no internal hardware failure, output within expected range)	
Crash control unit is connected	

▼B

Item	Technical principle applied for monitoring
No communication failures (bus connection failures) of relevant components in this table	
SIM is present (this item only applies if a removable SIM is used)	
Power source is connected	
Power source has sufficient charge (threshold at the discretion of the manufacturer)	

2. Test procedure

2.1. Self-test function verification test

▼M1

2.1.1. The following test shall be performed, separately for each of the items listed in the Table, on the vehicle with an eCall in-vehicle system installed in accordance with Article 5, on the STU in accordance with Article 7 or on the component, that is made part of a complete system for the purpose of the test, in accordance with Article 6.

2.1.2. Simulate a malfunction of the eCall system by introducing a critical failure monitored by the self-test function according to the technical documentation provided by the manufacturer. The manufacturer shall provide a list of the checks and a description of how to trigger them.

▼B

2.1.3. Power the eCall system up (e.g. by switching the ignition ‘on’ or activating the vehicle's master control switch, as applicable) and verify that the malfunction indicator illuminates shortly afterwards.

2.1.4. Power the eCall system down (e.g. by switching the ignition ‘off’ or deactivating the vehicle's master control switch, as applicable) and restore it to normal operation.

2.1.5. Power the eCall system up and verify that the malfunction indicator does not illuminate or extinguishes shortly after illuminating initially. ►M1 For the failures which cannot be simulated or injected by the technical service, the manufactures shall provide a documentation describing the test procedure and the test results to the technical service. ◄

3. Modification of type of 112-based eCall in-vehicle system or STU

▼M1

3.1. When the manufacturer submits an application for revision or extension of an existing type-approval for the purpose of including an alternative GNSS antenna, electronic control unit, mobile network antenna and/or power source components, no retesting of 112-based eCall in-vehicle system components shall be required for the purpose of fulfilling the requirements of this Annex, provided that those type-approved components possess at least the same functional features and that they are covered by this Annex in accordance with Article 6(3).

4. Technical requirements to enable periodic roadworthiness tests

4.1. Purpose

The purpose shall be to verify the following features of the eCall system:

▼M1

(a) its correct operational status, by visual observation of the failure warning signal status following the activation of the vehicle master control switch and any bulb check. Where the failure warning signal is only displayed in a common space (the area on which two or more information functions or symbols may be displayed, but not simultaneously), it must be checked first that the common space is functional prior to the failure warning signal status check;

(b) the correct accuracy of the Minimum Set of Data (by generating and reading current MSD), the correct function and condition of the eCall components and the backup-battery (if applicable), by the use of an electronic vehicle interface;

(c) the software integrity, by external verification of version information, hash values and system configurations against reference data;

(d) the correct functionality of the voice communication, by performing an audio echo and speaker test using the vehicle interface.

4.2. Requirements

4.2.1. The 112-based eCall in-vehicle system shall be able to provide the information to perform the methods of testing specified in Section 3, point 7.13, of Annex I to Directive 2014/45/EU using the electronic vehicle interface.

4.2.2. The manufacturer shall make available the technical information, which shall contain the instructions for reading out the information or performing the checks related to each item specified in Section 3, point 7.13, of Annex I to Directive 2014/45/EU.

4.3. Test procedure

4.3.1. It shall be verified that the information related to each item specified in Section 3, point 7.13 of Annex I to Directive 2014/45/EU, can be read out from the eCall system using the electronic vehicle interface according to the instructions of the manufacturer.

4.3.2. It shall be verified that an echo and speaker test can be performed to check the correct functionality of the voice communication via the vehicle interface.

▼B

ANNEX VIII

Technical requirements and test procedures related to privacy and data protection

PART I

Procedure for verifying the lack of traceability of an eCall in-vehicle system or STU

1. Purpose

1.1. This test procedure is to ensure that a 112-based eCall in-vehicle system or STU is not traceable and is not subject to any constant tracking in its normal operational status.

2. Requirements

▼M1

2.1. The 112-based eCall in-vehicle system or STU is not available for communication with the PSAP or at least does not automatically respond if the PSAP initiates the communication after expiry of the eCall timer T9 (1 hour).

---

3. Test conditions

▼B

3.1. The following tests shall be performed on a representative arrangement of parts (without a vehicle body).

▼M1

3.2. Before performing the test, ensure that:

(a) one of the connection procedures specified in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) the vehicle ignition or master control switch is activated;

(d) any TPS or added-value service system is disabled.

---

4. Test method

4.1. Perform a test eCall by applying a manual trigger of the system.

▼M1

4.2. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received a call or by a successful voice connection to the PSAP test point.

4.3. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

4.4. Leave the 112-based eCall IVS switched on and wait for at least 63 min (1 hour + 5 % margin according to EN 16454 and EN 17240).

4.5. Via the PSAP test point, attempt to connect to the 112-based eCall IVS.

5. Assessment

5.1. The requirement is determined to have been passed if the 112-based eCall in-vehicle system is neither available for communication with the PSAP nor automatically responds to the call when the PSAP test point attempts to connect after expiry of the eCall timer T9 (1 hour).

5.2. The establishment of connection with the 112-based eCall IVS or the automatic answering to the call when the PSAP test point initiates the communication constitute a failure.

▼B

PART II

Procedure for verifying the length of time an eCall log file is stored by the eCall in-vehicle system or STU

1. Purpose

1.1. This test procedure aims to ensure that personal data processed pursuant to Regulation (EU) 2015/758 is not retained by the eCall in-vehicle system longer than necessary for the purpose of handling the emergency situation and is fully deleted as soon as no longer necessary for that purpose.

1.2. This is to demonstrate the automatic deletion by proving that eCall log files are not kept beyond 13 hours from the point of initiating an eCall.

2. Requirements

2.1. When interrogated, the eCall in-vehicle system or STU shall not maintain any record of an eCall in its memory beyond 13 hours from the point of initiating an eCall.

3. Test conditions

3.1. The Technical Service shall be facilitated to have access to the part of the system where the eCall log files are stored in the IVS.

3.2. The following test shall be performed on a representative arrangement of parts.

▼M1

3.3. Before performing the test, ensure that:

▼M1

(a) one of the connection procedures defined in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) the vehicle ignition or master control switch is activated; and

(d) any TPS or added-value service system is disabled.

▼B

4. Test Method

▼M1

4.1. Perform an eCall by applying a manual trigger of the system.

4.2. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received a call or by a successful voice connection to the PSAP test point.

4.3. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

4.4. 13 hours after an eCall has been placed, the technical service tester shall be facilitated with access to where the eCall log files are stored in the IVS. This will involve the potential to download from the IVS any log files so that they can be viewed by the tester.

▼B

5. Assessment

5.1. The requirement is determined to have been passed if no log files are present in the eCall in-vehicle system memory.

5.2. The presence of a log file pertaining to an eCall that has occurred more than 13 hours ago constitutes a failure.

PART III

Procedure for verifying the automatic and continuous removal of data in the internal memory of an eCall in-vehicle system or STU

1. Purpose

1.1. This test procedure aims to ensure that personal data is only used for the purpose of handling the emergency situation and is automatically and continuously removed from the internal memory of the eCall in-vehicle system or STU.

1.2. This is to be proved by demonstrating that in the internal memory of the 112 based eCall in-vehicle system or STU, maximum of last three locations of the vehicle are retained.

▼B

2. Requirements

2.1. When interrogated, the eCall in-vehicle system or STU shall not maintain more than three recent locations of the vehicle.

3. Test conditions

▼M1

3.1. The technical service shall be facilitated to have access to the part of the system where the vehicle location data are stored in the IVS internal memory.

▼B

3.2. The following test shall be performed on a representative arrangement of parts.

▼M1

3.3. Before performing the test, ensure that:

(a) one of the connection procedures defined in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) the vehicle ignition or master control switch is activated; and

(d) any TPS or added-value service system is disabled.

▼B

4. Test Method

▼M1

4.1. Perform an eCall by applying a manual trigger of the system.

4.2. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received a call or by a successful voice connection to the PSAP test point.

4.3. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

4.4. The technical service tester shall be facilitated with access to where the vehicle location data are stored in the IVS internal memory. This will involve the potential to download from the IVS any stored locations so that they can be viewed by the tester.

▼B

5. Assessment

5.1. The requirement is determined to have been passed if maximum of last three locations are present in the eCall in-vehicle system memory.

5.2. The presence of more than three locations constitutes a failure.

▼B

PART IV

Procedure for verifying the non- exchange of personal data between an eCall in-vehicle system or STU and third party services systems

1. Purpose

1.1. This test procedure shall ensure that the 112-based eCall in-vehicle system or STU and any additional system functionality providing TPS eCall or an added-value service are designed in such a way that no exchange of personal data between them is possible at any time.

2. Requirements

2.1. The following requirements apply to eCall in-vehicle systems or STUs that shall be used in conjunction with a TPS eCall in-vehicle system functionality.

2.2. Performance requirements

2.2.1. There is no exchange of personal data between the 112-based eCall in-vehicle system or STU and any additional system functionality providing TPS eCall or an added-value service.

2.2.2. Following an eCall made via the 112-based eCall in-vehicle system or STU, no log of this eCall shall be recorded in the memory of the TPS eCall or added-value service system.

▼M1

3. Test conditions

▼B

3.1. The following tests shall be performed either on a vehicle with an eCall in-vehicle system installed or on a representative arrangement of parts.

▼M1

3.2. The TPS system shall be disabled for the duration of the eCall.

3.2.1. Before performing the test, ensure that:

(a) one of the connection procedures defined in point 2.7 of Annex I, as agreed between the technical service and the manufacturer, will be applied for any test eCall;

(b) the dedicated PSAP test point is available to receive an eCall emitted by the 112-based system;

(c) the vehicle ignition or master control switch is activated; and

(d) any TPS or added-value service system is disabled.

▼B

3.2.2. Perform a test call by applying a manual trigger of the system (push mode) with the TPS disabled.

▼B

3.2.3. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received a call initiation signal or by a successful voice connection to the PSAP test point.

3.2.4. Clear down the test call using the appropriate PSAP test point command (e.g. hang up).

3.2.5. If the call attempt of the 112-based system fails during the test, the test procedure may be repeated.

3.3. The lack of a log file in the TPS system shall be verified via access to the part of the system where eCall log files are stored.

3.3.1. The Technical Service tester shall be facilitated with access to where the eCall log files are stored in the IVS. This will involve the potential to download from the IVS any log files so that they can be viewed by the tester.

3.3.2. The requirement is determined to have been passed if no log files are present in the TPS system in-vehicle system memory.

3.3.3. The presence of a log file in the TPS system pertaining to an eCall that has occurred via the 112-based system constitutes a failure.

▼M1

————————

4. Test method

4.1. Perform a test eCall by applying a manual trigger of the system.

4.2. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received a call or by a successful voice connection to the PSAP test point.

4.3. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

4.4. If the call attempt of the 112-based system fails during the test, the test procedure may be repeated.

4.5. The technical service tester shall be facilitated with access to where the eCall log files are stored in the IVS. This will involve the potential to download from the IVS any log files so that they can be viewed by the tester.

4.6. The lack of a log file in the TPS system shall be verified via access to the part of the system where eCall log files are stored.

▼**M1**

5. Assessment

5.1. The requirement is determined to have been passed if no log files are present in the TPS system in-vehicle system memory.

5.2. The presence of a log file in the TPS system pertaining to an eCall that has occurred via the 112-based system constitutes a failure.

▼B

ANNEX IX

Classes of vehicles referred to in Article 2

Armoured vehicles of categories M_1 and N_1, as defined in point 5.2 of Part A of Annex II to Directive 2007/46/EC, equipped with armoured security glazing class BR 7 according to the classification under European standard EN 1063:2000 (Test and Classification for Ballistic Security Glazing) and with body parts complying with European standard EN 1522:1999 (Bullet Resistance in Windows, Doors, Shutters and Blinds), where those vehicles, due to their special purpose, cannot meet the requirements of Regulation (EU) 2015/758 and of this Regulation.

▼M1

ANNEX X

Test procedure for the verification of the performance of the back-up power supply

1. Purpose

The purpose of this test is to make sure that the eCall in-vehicle system or eCall STU is capable to communicate for the period specified in point 2.

2. Requirements

The eCall system or eCall STU shall be operable for a period of at least 5 minutes in voice communication mode followed by 60 minutes in call-back mode (idle mode, registered in the network) followed by another period of at least 5 minutes in voice communication mode.

3. Test conditions

The following verification test shall be performed on an eCall STU that has been subjected to the high-severity deceleration test according to Annex I.

If the eCall STU does not include the microphone(s) and speaker(s) for the eCall system, then representative microphone(s) and speaker(s) shall be added to the test setup in order to execute the test from this Annex.

3.1. Test method

3.1.1. Perform an eCall by applying a manual trigger of the system.

3.1.2. Verify that a call was established with the PSAP test point by a record of the PSAP test point showing that it received the call or by a successful voice connection to the PSAP test point.

3.1.3. Disconnect the main power source.

3.1.4. Read out any text for at least 5 minutes at the PSAP test point.

3.1.5. Clear down the eCall using the appropriate PSAP test point command (e.g. hang up).

3.1.6. Wait for 56 minutes after the call was ended.

3.1.7. Initiate a call from the PSAP test point to the eCall in-vehicle system.

3.1.8. If the call is automatically accepted, read out any text for at least 5 minutes at the PSAP test point, otherwise the test is finished.

▼**M1**

3.2. Assessment

3.2.1. The requirement is determined to have been passed if the eCall STU is capable to communicate for the required period, as specified in point 2.

3.2.2. The incapacity of the 112-based eCall in-vehicle system to communicate for the period referred to in point 2 constitutes a failure.